Course Name: Engineering Workshop Practice
Course Code: ME-102

Contact Hours/Week: 1L,2P **Course Credits:** 2

Course Objectives

- To study the fundamentals and have practical exposure of basic manufacturing processes.
- To familiarize the students with basics of machining, welding, fitting, smithy, carpentry, foundry and sheet metal related operations and handling/working of equipments & processes.
- To familiarize students with various handling/working of Soldering and Brazing processes.

Unit Number	Course Content	Contact Hours
UNIT-01	Importance of Engineering Workshop and description about each shop. Basic brief introduction of Engineering Materials and their classification. Introduction to Smithy shop, brief introduction of tools used in Smithy.	3L
UNIT-02	Brief description of machining operations. Lathe type and specifications. Study about construction and working of Lathe machine and operations.Brief description of various joining processes. Brief description about Arc Welding and Gas Welding processes. Introduction of Soldering & Brazing processes	3L
UNIT-03	Brief description about tools used in Foundry shop and methods of preparation of green sand mould, sand type and properties. Brief description of various fitting operations such as chipping, fitting. Scraping. Drilling, reaming & thread cutting. Function and classification of various hand tools used in fitting.	3L
UNIT-04	Classification of wood, importance of seasoning. Brief introduction of Carpentry tools. Different Carpentry joints & their application. Introduction to Sheet metal shop. Acquaintance with Sheet metal tools and their uses.	3L

List of Experiments

1. Preparation of job as per given drawing using Lathe Machine covering various operations using Lathe and attachment.
2. Preparation of job as per given drawing T Joint using Arc welding Setup, Soldering & Brazing Job Practice
3. Preparation of Green Sand Mould using tools of Foundry Shop by various method.
4. Preparation of job as per drawing for cutting practice and Dovetail Joint in Carpentry Shop using Carpentry Tool.
5. Preparation of job as per given drawing related to Fitting shop, and Preparation of job in sheet metal shop for Riveting operation.
6. Preparation of job as per given drawing (Chisel) using tools of Smithy Shop.

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Learn the basics of metal machining, fitting, forging, foundry and sheet metal related operations.
CO2: Develop the skill for soldering and brazing operations
CO3: Able to learn basic repair and maintenance techniques using welding processes.
CO4: Able to learn basic carpentry processes.

Text Books

1. A Course in Workshop Technology by B.S. Raghuwanshi, DhanpatRai &Company(P) Limited.
2. Elements of Workshop Technology by Hajra Choudhary & Nirjhar Roy, Media Promoters and PublishersPvt. Ltd.
3. Workshop Technology by R.S. Khurmi & J.K. Gupta, S.Chand

Reference Books

1. Mechanical Workshop Practice, by K. C. JOHN, PHI Learning Pvt. Ltd., 27-Aug-2010
2. Workshop Practice, by H.S. Bawa, McGraw Hill Education; 2nd edition
3. Manufacturing Practices (Workshop Practice), by Dr. R.K. Singal, S.K. Kataria & Sons